Spectral bounds and asymptotics of Casimir sources

Abstract

We study the finite Casimir face polynomial and the coefficients obtained by identifying its variables across a diagram. First we prove uniform exponential upper bounds for its central and one-sided degree sectors, over arbitrary face phases and the entire real spectral interval. The bounds are expressed by an explicitly normalized triangle variational problem. For a marked triangulated sphere with stationary positive face radii, we then prove a nonzero holomorphic Gaussian asymptotic for the shared central coefficient, together with a strict spectral gap away from zero. The proof combines exact cofactor algebra, signed transfer estimates, local analytic summation, and positive-matrix bounds on the other face phases. All integer levels, both parities, and variables fixed at zero are included. These results supply analytic inputs to a filling theorem; they do not estimate the full complementary source or identify the limiting action with a hyperbolic volume.

1 Local polynomials and the two asymptotic questions

A Casimir source is a homogeneous polynomial obtained from a finite matrix operator associated with a pair of strands. A diagram combines these local polynomials by identifying their face variables and extracting one coefficient. There are two different analytic questions. First, how large can a local polynomial be over all face phases and the full real spectral interval? Second, does the shared coefficient have a nonzero leading term, and can one bound the rest of its spectral domain?

We answer the first question for each of the three degree sectors of the local polynomial. For the second question we select the central sector at every matched edge of a triangulated sphere and assume that the associated triangle variational problem has stationary positive radii. Theorem 1.1 gives the local upper bounds; Theorem 8.1 gives a holomorphic Gaussian asymptotic for the shared central coefficient and a strict spectral gap away from zero. Their hypotheses and conclusions are kept separate. In particular, the second theorem does not estimate the complementary degree sectors.

The proofs start with exact finite cofactor identities. Signed transfer matrices and logarithmic integrals give the uniform local bounds. For the nonzero asymptotic, Euler–Maclaurin summation with Cauchy estimates constructs local analytic actions, and a contour deformation selects their contributing Gaussian. Positive finite matrices then control the other face phases before a common coefficient is extracted. The one-variable summation and saddle formulas are recorded in [3, §2.10(i), §2.4(iv)]; the uniform errors, contour contribution, and shared multivariable Gaussian are proved below.

The variables of a local face polynomial have a prescribed order: x, y are the outer variables, b the input middle variable, and c the output middle variable. Variables shared by different boxes must later be identified. This paper estimates one actual polynomial before that shared extraction. The estimates hold for all phases of the four variables, rather than only at their positive real values.

Fix an integer $N \geq 3$, and put

$$d = N - 1, \quad m = \lfloor N/2 \rfloor, \quad J = \lfloor d/2 \rfloor, \quad \theta = \pi/N, \quad q = e^{i\theta}, \quad Q = q^2.$$

The finite Fourier theorem in [1, Proposition 4.1 and equations (4.4)–(4.5)] identifies the physical pair Casimir Ω, one N-dimensional residue block at a time, with the polynomial operator specified in Section 2. The identification uses a finite Fourier transform, a transpose and a diagonal change of basis; it is stated precisely below. It also proves $\chi_N(\Omega) = 0$, where $\chi_N(z) = 2T_N(z/2) + 2$ and $T_N(\cos t) = \cos Nt$. The full resolvent numerator is the polynomial

$$V_N(\sigma; \Omega) = \frac{\chi_N(\sigma) - \chi_N(\Omega)}{\sigma - \Omega}.$$

The fraction is polynomial division, not an inverse at a spectral root. Its degree in σ is d; it retains the first derivatives at double roots as well as the values at distinct roots.

Write $H_V(x, b, c, y; \sigma)$ for its homogeneous face polynomial. Section 2 gives a division-free definition and fixes its sign. A decreasing path from row j to row i contributes the middle monomial $b^i c^{d-j}$. When $N = 2m$, its central sector has $i \leq m - 1, j \geq m$; when $N = 2m + 1$, it has $i \leq m, j \geq m$. The left sector has $j < m$; the right consists of the remaining paths above the central cut. Denote these three polynomials by $H_{V,\mathrm{C}}, H_{V,\mathrm{L}}, H_{V,\mathrm{R}}$. This is a partition of the full numerator paths, not a new spectral operator.

Set

$$\mathcal{B}(t) = -\int_0^t \log|2 \sin v| \, dv, \qquad e(a) = \frac{\mathcal{B}(\pi a) + \mathcal{B}(\pi/2 - \pi a)}{\pi}, \quad 0 \leq a \leq \tfrac{1}{2}.$$

For nonnegative w_1, w_2, w_3, not all zero, define

$$\mathcal{Q}(w_1, w_2, w_3) = \max_{\substack{a_i \geq 0, \ \sum_i a_i = 1/2 \\ a_i = 0 \text{ if } w_i = 0}} \left\{ \sum_i e(a_i) + \sum_{w_i > 0} a_i \log w_i \right\}. \tag{1.1}$$

A variable fixed at zero has exactly zero load. It is never replaced silently by a small positive radius. The logarithmic integrals are finite at their endpoints. To fix their normalization, put $I = \int_0^{\pi/2} \log \sin t \, dt$. Reflection and the double-angle formula give $2I = I - (\pi/2) \log 2$. Hence $\mathcal{B}(0) = \mathcal{B}(\pi/2) = 0$, and $e(0) = e(1/2) = 0$.

In the theorem the lowercase letters denote nonnegative moduli. For $x + y > 0$, put

$$\begin{aligned} &Q_b = \mathcal{Q}(x, b, y), && Q_c = \mathcal{Q}(x, c, y), \quad X = Q_b + Q_c, \\ &A_{\mathrm{L}} = \max\{\log c, Q_b + \tfrac{1}{2} \log c\} && (c > 0), \\ &A_{\mathrm{R}} = \max\{\log b, Q_c + \tfrac{1}{2} \log b\} && (b > 0). \end{aligned} \tag{1.2}$$

If $c = 0$, the left polynomial is zero and its rate is interpreted as $-\infty$; the same convention applies on the right when $b = 0$.

Theorem 1.1 (Uniform local sector bounds)**.** *Fix nonnegative face moduli (x, b, c, y) with $x + y > 0$. For $s = \mathrm{C}, \mathrm{L}, \mathrm{R}$,*

$$\limsup_{N \to \infty} \frac{1}{N} \log \sup_{\substack{\sigma \in [-2,2] \\ |X_0| = x, \ |B_0| = b, \ |C_0| = c, \ |Y_0| = y}} |H_{V,s}(X_0, B_0, C_0, Y_0; \sigma)| \leq \begin{cases} X & s = \mathrm{C}, \\ A_{\mathrm{L}} & s = \mathrm{L}, \\ A_{\mathrm{R}} & s = \mathrm{R}. \end{cases} \tag{1.3}$$

The limit superior is over all integer levels, including both parities and composite levels. The statement is uniform when the zero pattern is fixed and every nonzero modulus varies

in a fixed compact subset of $(0,\infty)$*. Thus, for every* $\varepsilon>0$*, there is one threshold* N_0 *on such a compact set for which the indicated rate plus* ε *bounds the normalized logarithm at every* $N\ge N_0$*.*

The supremum precedes the limit, so sampled spectral zeros are included. The proof never divides a finite numerator by such a zero, and never replaces a negative tail coefficient by its absolute value inside the recurrence.

The admissible vertex with load $1/2$ on b gives

$$\mathcal{Q}(x,b,y)\ge \tfrac{1}{2}\log b \quad (b>0), \qquad \max\{X,A_{\mathrm{L}},A_{\mathrm{R}}\}=\max\{X,\log b,\log c\}, \tag{1.4}$$

with undefined logarithms omitted. Each mixed expression in (1.2) is at most X. This bounds the full local polynomial. A shared coefficient must still be estimated at one common set of face radii, because the same variable can occur in several local factors.

2 The exact polynomial and its path partition

We import only the finite operator identification from [1]. The matrix needed in the estimates is explicit. On $1,z,\dots,z^d$, let

$$\mathsf{C}z^j=2\cos((2j+1)\theta)z^j-s_j^2(x+y)z^{j-1}-s_js_{j-1}xy\,z^{j-2}, \qquad s_j=2\sin(j\theta). \tag{2.1}$$

Negative powers vanish because $s_0=0$. In the finite Fourier theorem $\mathsf{C}=-T$. To specify the imported identification, let Ω_s be the block of the physical pair operator on

$$e_u\otimes e_{[s-u]_N}, \qquad 0\le u<N, \qquad 0\le s<N.$$

Here $e_0,\dots,e_d$ is its standard one-strand basis, and $[v]_N$ is the representative in $\{0,\dots,N-1\}$. For $y\neq 0$, specialize $x=Q^{s+1}y$, set $\zeta_u=-q^{-1}xQ^{-u}$, and define $\mathsf{E}_{uj}=\zeta_u^j$, $G_{uu}=q^{du}$. Then

$$\mathsf{E}\mathsf{C}\mathsf{E}^{-1}=G^{-1}\Omega_s^{\mathsf{T}}G. \tag{2.2}$$

The N blocks form the N^2-dimensional pair space. The identity includes their nilpotent parts, not just eigenvalues. The face polynomial retains independent x,y; its specialized values recover the matrix entries by the adapter in [1, equation (4.5)]. If $f(\mathsf{C})_{ij}$ is the coefficient from z^j to z^i, the homogeneous polynomial is

$$H_f(x,b,c,y)=\sum_{0\le i\le j\le d} f(\mathsf{C})_{ij}b^ic^{d-j}.$$

Its entries are polynomials in x,y, and its total face degree is d.

Set

$$\sigma=-2-u, \qquad a_j=2+2\cos((2j+1)\theta), \qquad \alpha_j=a_j+u, \qquad A=x+y, \quad P=xy. \tag{2.3}$$

The triangular matrix $\mathsf{M}=2I+\mathsf{C}+uI$ has diagonal α_j, and lowering entries $-s_j^2A$, $-s_js_{j-1}P$. Its diagonal product is $(-1)^N\chi_N(-2-u)$. Consequently

$$\mathcal{P}_N(x,b,c,y;u):=(-1)^dH_V(x,b,c,y;-2-u)=\sum_{i\le j}(\operatorname{adj}\mathsf{M})_{ij}b^ic^{d-j}. \tag{2.4}$$

The identity follows first where M is invertible and then by polynomial equality, including every singular spectral value.

Expand its triangular inverse by the strictly lowering part and multiply by all diagonal factors. A path $j \to i$ uses steps of length one or two. Its weight is the product of $s_r^2 A$ for a step $r \to r-1$, $s_r s_{r-1} P$ for a step $r \to r-2$, and α_r for precisely the rows it does not visit. The middle monomial $b^i c^{d-j}$ completes its term. This proves the path rule, including the case of a zero unused factor.

Partition $\mathcal{P}_N = \mathcal{P}_{\mathrm{C}} + \mathcal{P}_{\mathrm{L}} + \mathcal{P}_{\mathrm{R}}$ by the sectors in the introduction. Their common sign relative to H_V does not affect modulus bounds. Every sector has spectral degree at most d. At $u = 0$, every path weight is nonnegative. At odd N, a path missing the middle row has its unused factor $a_m = 0$.

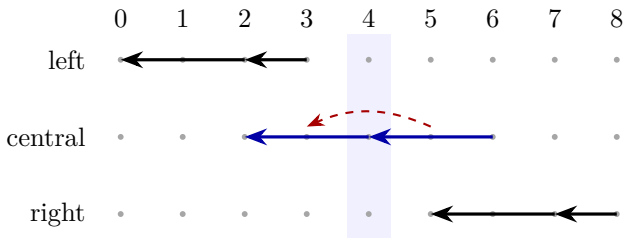


Figure 1: The partition at the illustrative odd level $N = 9$. A central path may visit row m, or skip it by $m+1 \to m-1$. The skipped row contributes its unused factor u, retained in the exact central polynomial even though it vanishes at $u = 0$.

2.1 Cleared cofactors

Let $R_j(b;u)$ be the determinant obtained from the block of M on rows $0, \dots, j$ by replacing its last row with $(1, b, \dots, b^j)$. Its path expansion gives

$$\begin{aligned} R_0 = 1, \qquad R_{-1} = R_{-2} = 0, \\ R_j(b;u) = b^j \prod_{r<j} \alpha_r + s_j^2 A R_{j-1}(b;u) + s_j s_{j-1} \alpha_{j-1} P R_{j-2}(b;u). \end{aligned} \tag{2.5}$$

Equivalently, group the paths by their starting row:

$$\begin{aligned} R_j = \sum_{i=0}^{j} b^i \prod_{r<i} \alpha_r K_{i,j}, \qquad K_{i,i} = 1, \quad K_{i,i-1} = 0, \\ K_{i,k} = s_k^2 A K_{i,k-1} + s_k s_{k-1} \alpha_{k-1} P K_{i,k-2}. \end{aligned} \tag{2.6}$$

The interval polynomial $K_{i,j}$ sums the paths between rows i and j. Both this formula and (2.5) are polynomial identities when a diagonal factor vanishes.

The first term at $j = 0$ is the empty product one. Each R_j has face degree j and spectral degree at most j. Put

$$D_b = R_{m-1}(b;u), \qquad E_b = \alpha_{m-1} R_{m-2}(b;u),$$

with $E_b = 0$ when $m = 1$. Subscripts b, c here denote substitution of the middle variable, not differentiation.

Lemma 2.1 (Exact central factors)**.** *At $N = 2m$,*

$$\mathcal{P}_{\mathrm{C}} = s_m^2 A D_b D_c + s_m s_{m-1} P (D_b E_c + E_b D_c). \tag{2.7}$$

At $N = 2m + 1$, with $G_b = R_m(b; u)$,

$$\mathcal{P}_{\mathrm{C}} = G_b G_c + u s_m^2 P D_b D_c. \tag{2.8}$$

Both are polynomial identities for every u.

Proof. An even central path crosses the cut exactly once, by $m \to m-1$, $m+1 \to m-1$, or $m \to m-2$. Their weights give the three terms. Reflection $r \mapsto d-r$ identifies upper and lower cofactors: $a_{d-r} = a_r$, and the reflected step products agree. At odd level, paths visiting m split there to give $G_b G_c$. A path skipping it has the unique jump $m+1 \to m-1$, with weight $s_{m+1} s_m P = s_m^2 P$; its unused diagonal is u. This proves the second term without dividing by u. □

The same reflection interchanges the left and right sectors and b, c. A one-step at r reflects to one at $N - r$, with $s_{N-r} = s_r$; a two-step reflects to one at $N + 1 - r$, with product $s_{r-1} s_r$. It therefore suffices to estimate the left sector.

3 Exact nodal control of the central sector

For face polynomials, $F \preccurlyeq G$ means that every coefficient of $G - F$ is nonnegative real. The symbol $|F|_{\mathrm{coef}}$ replaces each coefficient by its modulus. Recall that $J = \lfloor (N-1)/2 \rfloor$. The $J + 1$ distinct physical nodes in the u coordinate are

$$u_a = -a_a, \qquad 0 \le a \le J.$$

They include zero at odd level, but not at even level.

Lemma 3.1 (Nodal positivity and comparison)**.** *For every $0 \le j, a \le J$,*

$$0 \preccurlyeq R_j(b; u_a) \preccurlyeq R_j(b; 0).$$

Proof. If $j \le a$, every diagonal factor used in (2.5) lies between zero and its value at $u = 0$; induction proves the claim. If $j > a$, the inhomogeneous term vanishes after row a. Every surviving path must visit row a, so

$$R_j(b; u_a) = K_{a,j}(u_a) R_a(b; u_a),$$

with $K_{a,j}$ defined in (2.6). Normalize $q_r = K_{a,a+r} / \prod_{v=a+1}^{a+r} s_v^2$. For $r \ge 2$, $j = a + r \le J$, product-to-sum gives

$$a_{j-1} - a_a = -s_{j+a} s_{j-a-1}, \qquad s_j s_{j-1} - s_{j+a} s_{j-a-1} = s_a s_{a+1}.$$

All the indicated sines are nonnegative. Thus

$$q_0 = 1, \quad q_1 = x + y, \quad q_r = (x+y) q_{r-1} - \gamma_r x y q_{r-2}, \qquad \gamma_r = 1 - \frac{s_a s_{a+1}}{s_j s_{j-1}} \in [0, 1]. \tag{3.1}$$

The identity

$$q_r - x q_{r-1} = y(q_{r-1} - x q_{r-2}) + (1 - \gamma_r) x y q_{r-2}$$

proves simultaneous coefficientwise positivity, starting from $q_0 = 1, q_1 - x q_0 = y$. Also $q_r \preccurlyeq (x+y)^r$. At $u = 0$, the interval continuant contains this all-one-step term and other nonnegative terms. Hence $K_{a,j}(u_a) \preccurlyeq K_{a,j}(0)$. The product $K_{a,j}(0) R_a(b; 0)$ is the subset of paths in $R_j(b; 0)$ whose start is at or below a and which visit a. This proves the assertion without assuming positivity of the signed recurrence. □

Lemma 3.2 (Exact interpolation bound)**.** *An absolute constant C gives, for $j \le J$,*

$$|R_j(b;u)|_{\mathrm{coef}} \preccurlyeq C(1+\log N)R_j(b;0), \qquad -4 \le u \le 0. \tag{3.2}$$

Proof. Each face coefficient is a polynomial in u of degree at most $j \le J$. This interpolation is not applied to the degree-d full resolvent. Use $z = -1 - u/2$ and $\phi_k = (2k+1)\pi/N$, $0 \le k < N$. For a polynomial $p(z)$ of degree at most J, discrete Fourier inversion gives

$$p(\cos t) = \frac{1}{N}\sum_{k=0}^{N-1} D_J(t-\phi_k)p(\cos\phi_k), \qquad D_J(v) = \sum_{\ell=-J}^{J} e^{i\ell v}.$$

The difference of any two retained modes has magnitude less than N, so its grid average is a Kronecker delta. Pairing equal cosine nodes gives ordinary interpolation, with the unpaired point -1 at odd N. Since

$$|D_J(v)| \le \min(2J+1, |\sin(v/2)|^{-1}),$$

the normalized sum of absolute kernel values on any translate of the grid is at most $C(1+\log N)$. The two nearest points cost a constant; successive distances k/N cost at most C/k. Pairing cannot increase this bound. Apply it to each coefficient and use Lemma 3.1. □

Proposition 3.3 (Central polynomial comparison)**.** *For an absolute C and every $N \ge 3$,*

$$|\mathcal{P}_{\mathrm{C}}(u)|_{\mathrm{coef}} \preccurlyeq C(1+\log N)^2\mathcal{P}_{\mathrm{C}}(0), \qquad -4 \le u \le 0. \tag{3.3}$$

Proof. Apply (3.2) to the exact central factors. The scalar factors $|u|, |a_{m-1}+u|$ are at most four. At even $N \ge 4$,

$$D_b(0) \succcurlyeq s_{m-1}^2(x+y)R_{m-2}(b;0), \qquad (x+y)^2 \succcurlyeq 2xy.$$

The sine is uniformly bounded below. Thus either mixed term $xyD_b(0)R_{m-2}(c;0)$ is bounded coefficientwise by a constant times $(x+y)D_b(0)D_c(0)$, already present at zero. At odd levels,

$$G_b(0) \succcurlyeq s_m^2(x+y)D_b(0), \qquad G_b(0)G_c(0) \succcurlyeq 2s_m^4 xyD_b(0)D_c(0).$$

Here s_m is bounded below, including $N = 3$. The skipped-middle term is therefore controlled as well. □

4 A signed interval transfer estimate

This section bounds the signed recurrence. It is not a Perron estimate for its entrywise absolute value. Parametrize the real interval by

$$\sigma = -2\cos(2\pi\delta), \qquad u = -4\sin^2(\pi\delta), \qquad 0 \le \delta \le \tfrac{1}{2}.$$

For positive outer sum $A = x+y$ and $P = xy \ge 0$, define

$$\begin{aligned} a_\delta(t) &= 4(\cos^2(\pi t) - \sin^2(\pi\delta)), \\ \rho_\delta(t) &= A\sin(\pi t) + \sqrt{(x-y)^2\sin^2(\pi t) + 4P\cos^2(\pi\delta)}, \\ \lambda_\delta(t) &= 2\sin(\pi t)\rho_\delta(t). \end{aligned} \tag{4.1}$$

On each compact set of allowed moduli,

$$ct \le \rho_\delta(t) \le C, \qquad ct^2 \le \lambda_\delta(t) \le C, \qquad 0 < t \le \tfrac{1}{2}, \tag{4.2}$$

uniformly in δ.

Lemma 4.1 (Uniform signed transfer)**.** *For every $\varepsilon > 0$, all sufficiently large N satisfy*

$$|K_{i,j}| \le \exp\left\{ N \int_{i/N}^{j/N} \log \lambda_\delta(t)\, dt + \varepsilon N \right\}, \qquad 0 \le i \le j \le m. \tag{4.3}$$

The threshold is uniform in endpoints, real spectral parameter, and each fixed compact set of allowed moduli.

Proof. Normalize $L_{i,j} = K_{i,j} / \prod_{r=i+1}^{j} s_r$. Its step matrix is

$$T_h(t,\delta) = \begin{pmatrix} 2A\sin(\pi t) & 4P[\cos^2(\pi(t-h/2)) - \sin^2(\pi\delta)] \\ 1 & 0 \end{pmatrix}, \quad h = N^{-1}, \quad t = jh.$$

The limiting matrix has real eigenvalues

$$A\sin(\pi t) \;\pm\; \sqrt{(x-y)^2 \sin^2(\pi t) + 4P\cos^2(\pi\delta)}.$$

Its spectral radius is $\rho_\delta(t)$, including coincident roots. Use the Euclidean operator norm. Fix $\eta > 0$ and restrict to $t \in [\eta, 1/2]$, where $\rho_\delta(t) \ge c_\eta > 0$. Cayley–Hamilton gives uniformly

$$\|T^k\| \le C_\eta (k+1) \rho(T)^k. \tag{4.4}$$

For distinct eigenvalues the power divided difference is a sum of k terms, each bounded by ρ^{k-1}. The polynomial identity extends to coincident eigenvalues by its derivative value. Bounded entries and the positive lower bound on ρ give C_η. No eigenbasis or its condition number is used.

Choose a fixed block length k for which $\log(2C_\eta(k+1))/k$ is smaller than a prescribed error. Within a block, the exact matrices differ from its first limiting matrix by $O_k(N^{-1})$. Expanding the finite product one factor at a time bounds its difference from T^k by the same order. For large N this is at most $C_\eta(k+1)\rho^k$. Multiplying block bounds and using uniform continuity of $\log \rho$ gives its integral plus the prescribed error. Incomplete blocks have at most $2k$ factors and cost $o(N)$, uniformly in the endpoints.

For $0 < t < \eta$, bounded entries cost at most $C^{\eta N + O(1)}$. The omitted logarithmic integral has absolute value at most $C\eta\log(2/\eta)$, by (4.2). The exact scalar factors s_r have the same integrable endpoint bound, and their logarithmic Riemann sums converge uniformly over subintervals. Choose η first, then the fixed block length, and finally N. Restoring these factors proves the estimate. If a perturbation of the reference matrices is wanted, it must be chosen after the block length; a sufficiently small fixed perturbation then fits the same finite-product error. No complex perturbation is needed for the theorem here. □

4.1 Unused spectral rows

The exact diagonals are $\alpha_r = a_\delta((r+1/2)/N)$. Uniformly in δ and interval endpoints,

$$\sum_{r=l}^{k-1} \log|\alpha_r| \le N \int_{l/N}^{k/N} \log|a_\delta(t)|\, dt + o(N). \tag{4.5}$$

A sampled zero makes the left side $-\infty$, which causes no upper-bound problem. This convention describes a zero product; no finite division by it is made.

For a proof, truncate the logarithm from below at $-M$. The truncated function is jointly continuous, so its Riemann sums converge uniformly. The factorization

$$a_\delta(t) = 4\cos(\pi(t+\delta))\cos(\pi(t-\delta))$$

shows that $\{|a_\delta| < r\}$ has length at most $C\sqrt{r}$, uniformly in δ. Integrating this distribution bounds the omitted logarithmic tails uniformly by a quantity tending to zero as $M \to \infty$. The argument applies to every subinterval.

There is also the exact complete-prefix identity

$$\int_0^{1/2} \log|a_\delta(t)|\,dt = 0. \tag{4.6}$$

Shift the two cosine integrals and use the reflection and period of $\log|2\cos(\pi t)|$. Together they have the integral of a full period, which is zero. Integrability includes $\delta = 0, 1/2$. We use this identity only after both actual unused products have been bounded separately.

5 The normalized triangle action

For $b > 0$, define

$$F_b(\mu,\delta) = \mu\log b + \int_0^{\mu} \log|a_\delta(t)|\,dt + \int_{\mu}^{1/2} \log\lambda_\delta(t)\,dt. \tag{5.1}$$

When $b = 0$, only $\mu = 0$ is allowed. Use the exact starting-row decomposition (2.6). Lemma 4.1, (4.5), and at most $N+1$ starts give, for $n = N/2 + O(1)$ and $n \le m$,

$$|R_n(b;u)| \le \exp\left\{N\left[\max_{0\le\mu\le1/2} F_b(\mu,\delta) + \varepsilon\right]\right\} \tag{5.2}$$

uniformly for large N. Replacing n/N by $1/2$ costs $o(N)$, since λ_δ is uniformly positive at $1/2$.

For $0 \le \delta < 1/2$, put $s = \sin(\pi\delta)$, $k = \cos(\pi\delta)$, and

$$T_b = \sqrt{bA + P + b^2s^2}, \qquad \mathcal{D} = A^2 - 4Ps^2, \qquad \mathcal{E}_b = A + 2bs^2 + 2sT_b.$$

These scalar symbols are distinct from the finite cofactors D_b, E_b. The stationary point before the sign change $t_c = 1/2 - \delta$ is

$$\mu_b = \frac{1}{\pi}\arctan\frac{bk}{T_b}, \qquad \sin^2(\pi\mu_b) = \frac{k^2b^2}{(b+x)(b+y)}. \tag{5.3}$$

For $b = 0$ set $\mu_b = 0$, and write $Q_b(\delta) = F_b(\mu_b,\delta)$.

Lemma 5.1 (Calibration and spectral derivative)**.** *For the allowed nonnegative weights,*

$$Q_b(0) = \mathcal{Q}(x,b,y). \tag{5.4}$$

For $0 < \delta < 1/2$,

$$Q_b'(\delta) = -\log(\mathcal{E}_b/\sqrt{\mathcal{D}}) \le 0. \tag{5.5}$$

If $b = P = 0$*, the action is the constant* $\frac{1}{2}\log A$*. The endpoint and zero-weight cases follow from the exact sole-start or one-step formulas and their continuous limits.*

Proof. The stationary equation is $b\,a_\delta(\mu) = \lambda_\delta(\mu)$ where $a_\delta > 0$. Squaring its positive sides gives (5.3).

First fix the constant in (5.4). For $0 < a < 1/2$,

$$e'(a) = \log\cot(\pi a), \qquad e''(a) = -\frac{\pi}{\sin(\pi a)\cos(\pi a)} < 0.$$

At positive weights the entropy maximizer is unique and has loads

$$a_w = \frac{1}{\pi}\arctan\frac{w}{\sqrt{xb+xy+by}}, \qquad w = x, b, y.$$

The tangent addition identity makes their sum $1/2$, and the three constrained derivatives are equal. Hence the derivative of $\mathcal{Q}(x, b, y)$ with respect to $\log b$ is $a_b = \mu_b|_{\delta=0}$. The envelope rule gives the same derivative for $F_b(\mu_b, 0)$, so their difference is independent of $b > 0$. As $b \to \infty$, both minus $\frac{1}{2}\log b$ tend to zero. For the entropy this follows also directly from the displayed loads: the other loads and their entropy terms tend to zero, and their products with $\log b$ vanish. For the integral, $1/2 - \mu_b = O(b^{-1/2})$, the full prefix integral is zero, and the remaining logarithmic tails vanish. This proves equality without a geometric normalization. Zero weights follow by continuity on the constrained faces of the load simplex and by the dominated logarithmic integrals.

For the derivative, $\partial_{\log b} Q_b = \mu_b$, and direct differentiation gives

$$\partial_\delta \mu_b = -\frac{bs}{T_b}, \qquad \partial_{\log b}\log(\mathcal{E}_b/\sqrt{\mathcal{D}}) = \frac{bs}{T_b}.$$

Thus $Q_b' + \log(\mathcal{E}_b/\sqrt{\mathcal{D}})$ is independent of b. Determine it at $b = 0$, retaining the integration constant. For $P > 0$, set $\lambda = 2\sqrt{P}/A \in (0, 1]$. Differentiating the sole-start integral and putting $\vartheta = \pi t$ gives

$$Q_0' = -\lambda^2 sk\int_0^{\pi/2} \frac{d\vartheta}{W(\vartheta)(\sin\vartheta + W(\vartheta))}, \quad W(\vartheta) = \sqrt{(1-\lambda^2)\sin^2\vartheta + \lambda^2 k^2}.$$

Rationalization and $z = \cos\vartheta$ turn this into

$$-sk\int_0^1 \left\{\frac{1}{\sqrt{1-z^2}} - \frac{1}{\sqrt{1-\lambda^2 s^2 - (1-\lambda^2)z^2}}\right\}\frac{dz}{z^2 - s^2}.$$

The apparent pole at $z = s$ is removable. For either term, $B - Cs^2 = k^2$, and differentiation verifies

$$\int \frac{dz}{(z^2-s^2)\sqrt{B-Cz^2}} = \frac{1}{2sk}\log\left|\frac{kz/\sqrt{B-Cz^2} - s}{kz/\sqrt{B-Cz^2} + s}\right|.$$

Subtract the antiderivatives before evaluating. Their singularities at $z = s$ cancel. At zero both are zero; at one the first tends to zero and the second gives the finite remaining term. The result is

$$Q_0' = -\operatorname{artanh}(\lambda s) = -\log\frac{A + 2s\sqrt{P}}{\sqrt{A^2 - 4Ps^2}}.$$

This proves (5.5) for all $b \geq 0$. When $P = 0$, the exact tail root is $4A\sin^2(\pi t)$; direct differentiation gives the same formula. It also follows by dominated convergence as $P \downarrow 0$. If $b = P = 0$, the integral is $\frac{1}{2}\log A$. Finally $\mathcal{E}_b \geq \sqrt{\mathcal{D}} > 0$ in the open interval. The possible singularity at $x = y, \delta = 1/2$ is logarithmic and integrable. □

Lemma 5.2 (The cofactor maximum)**.** *For $b > 0$,*

$$\max_{0\leq\mu\leq 1/2} F_b(\mu, \delta) = \max\{Q_b(\delta), \tfrac{1}{2}\log b\}.$$

For $b = 0$, the sole value is $Q_0(\delta)$.

Proof. At a regular point, $\partial_\mu F_b = \log(b|a_\delta|/\lambda_\delta)$. Writing $w = k^2/\sin^2(\pi\mu)$, the ratio is

$$\frac{2b|w-1|}{A+\sqrt{(x-y)^2+4Pw}}.$$

Before t_c, w decreases from infinity to one and this ratio strictly decreases from infinity to zero. To check its monotonicity for $w > 1$, the positive derivative numerator after cancellation is

$$A + \frac{(x-y)^2 + 2Pw + 2P}{\sqrt{(x-y)^2+4Pw}}.$$

The case $P = 0$ is immediate. Thus the unique maximum there is μ_b. After t_c the ratio increases with μ, so every critical point is a minimum. The last endpoint has value $\frac{1}{2}\log b$ by (4.6); the first is below the pre-sign-change maximum. At $\delta = 1/2$, the ratio is the constant $b/\max(x,y)$, and the endpoint values prove the assertion. The continuous limit is $Q_b(1/2) = \frac{1}{2}\log\max(x,y)$. □

At $\delta = 0$, these lemmas and $\mathcal{Q}(x,b,y) \geq \frac{1}{2}\log b$ imply

$$R_{N/2+O(1)}(b;0) \leq \exp\{N(\mathcal{Q}(x,b,y)+\varepsilon)\}.$$

Use this in the exact central factorization at zero. The scalar coefficients and face monomials are bounded on each fixed compact. Proposition 3.3 then proves the central part of Theorem 1.1, including arbitrary face phases.

6 The one-sided envelope with both unused products

For $0 \leq i \leq j < m$, the exact left summand is

$$A_{\mathrm{rest}}(u) b^i c^{d-j} \left(\prod_{r<i} \alpha_r\right) \left(\prod_{j<r<m} \alpha_r\right) K_{i,j}(u), \tag{6.1}$$

where $A_{\mathrm{rest}} = \prod_{r=m}^{d} \alpha_r$ is the whole omitted upper-half product. On the real interval it has no positive exponential cost. At $N = 2m$,

$$A_{\mathrm{rest}}(u) = 2(-1)^m T_m(\sigma/2), \qquad |A_{\mathrm{rest}}| \leq 2.$$

Its roots and leading coefficient prove the identity. At $N = 2m+1$, it is, up to sign, $(\sigma+2)P_m(\sigma)$, where

$$P_m(2\cos t) = \frac{\cos((m+1/2)t)}{\cos(t/2)}.$$

The quotient is a finite alternating sum of m paired exponentials and one constant, so $|P_m| \leq 2m+1 = N$. Hence $|A_{\mathrm{rest}}| \leq 4N$, including the removable endpoint. These are the original unused products, not auxiliary normalizations.

For $b, c > 0$, apply the transfer and Riemann estimates separately to the two outside products in (6.1). Only then use (4.6). With $\mu = i/N, \nu = j/N$, the upper action becomes

$$\begin{aligned} F_{\mathrm{L}}(\mu,\nu,\delta) &= \mu\log b + (1-\nu)\log c + \int_\mu^\nu \log L_\delta(t)\,dt, \\ L_\delta(t) &= \frac{\lambda_\delta(t)}{|a_\delta(t)|}, \qquad 0 \leq \mu \leq \nu \leq \tfrac{1}{2}. \end{aligned} \tag{6.2}$$

This quotient is only continuum notation. No finite division by α_r occurred. Moving an endpoint by $O(N^{-1})$ changes the integrals by $o(1)$ uniformly, by logarithmic integrability. There are at most N^2 pairs (i,j), contributing only $o(N)$. If $b=0$, the finite sum has only $i=0$, so only $\mu=0$ is allowed. If $c=0$, the left polynomial vanishes because $d-j\ge\lfloor N/2\rfloor>0$.

Proposition 6.1 (Exact continuum envelope)**.** *For $c>0, b\ge 0$,*

$$\sup_{\substack{0\le\delta\le 1/2\\ 0\le\mu\le\nu\le 1/2}} F_{\mathrm{L}}(\mu,\nu,\delta)=\max\{\log c,\mathcal{Q}(x,b,y)+\tfrac{1}{2}\log c\}. \tag{6.3}$$

For $b=0$ restrict the left side to $\mu=0$. This is an equality of continuum upper envelopes, not a lower bound for an oscillatory finite polynomial.

Proof. First assume $b>0$, $0<\delta<1/2$, and set $z=\sin^2(\pi\delta)$, $\mathcal{D}=A^2-4Pz$. The ratio calculation in Lemma 5.2 shows that L_δ increases from zero to infinity before t_c and decreases from infinity to

$$L_{\mathrm{end}}=\frac{A+\sqrt{\mathcal{D}}}{2z}$$

afterward. The endpoint derivatives are

$$\partial_\mu F_{\mathrm{L}}=\log b-\log L_\delta(\mu),\qquad \partial_\nu F_{\mathrm{L}}=\log L_\delta(\nu)-\log c.$$

A stationary $\nu<t_c$ is a minimum, and a stationary $\mu>t_c$ is a minimum. The logarithmic singularity creates no maximum: the derivative in ν is positive on both sides near t_c, and that in μ is negative. The boundary $\mu=0$ is excluded for $b>0,\nu>0$ by its positive initial derivative. Apart from the diagonal $\mu=\nu$, the possible maxima therefore have $\mu=\mu_b$ and either $\nu=1/2$ or a stationary $\nu_c>t_c$.

The diagonal contributes at most $\max(\log c,\frac{1}{2}\log b+\frac{1}{2}\log c)$. The terminal value is $Q_b(\delta)+\frac{1}{2}\log c$, because the complete prefix integral is zero. Both are bounded by the proposed answer, using Lemma 5.1 and $\mathcal{Q}(x,b,y)\ge\frac{1}{2}\log b$.

A post-change stationary endpoint exists precisely when $c\ge L_{\mathrm{end}}$. If $c\le\max(x,y)$, this never occurs in the open interval. Otherwise it begins at δ_c, where

$$\sin^2(\pi\delta_c)=\frac{A}{c}-\frac{P}{c^2},\qquad \sin^2(\pi\nu_c)=\frac{c^2\cos^2(\pi\delta)}{(c-x)(c-y)}. \tag{6.4}$$

At entry $\nu_c=1/2$, and afterward $t_c<\nu_c<1/2$. Define

$$J_c(\delta)=\nu_c\log c+\int_0^{\nu_c}\log|a_\delta(t)|\,dt+\int_{\nu_c}^{1/2}\log\lambda_\delta(t)\,dt.$$

The straddling value is exactly

$$S_{\mathrm{L}}=Q_b(\delta)+\log c-J_c(\delta). \tag{6.5}$$

Subtracting the displayed integrals and using (4.6) proves this with no additive constant.

Put $T_c^-=\sqrt{c^2z-cA+P}$ and $\mathcal{H}_c=2cz-A+2\sqrt{z}\,T_c^-$. On the strict branch,

$$J_c'=-\log(\mathcal{H}_c/\sqrt{\mathcal{D}}). \tag{6.6}$$

Here is a normalization proof. The stationary envelope rule gives $\partial_{\log c}J_c=\nu_c$, while direct differentiation gives

$$\partial_\delta\nu_c=-\frac{c\sin(\pi\delta)}{T_c^-},\qquad \partial_{\log c}\log\mathcal{H}_c=\frac{c\sin(\pi\delta)}{T_c^-}.$$

Thus the difference in (6.6) is independent of c for each δ. At $c \downarrow L_{\mathrm{end}}$, one has $\nu_c \to 1/2$, $T_c^- \to 0$, and $\mathcal{H}_c \to \sqrt{\mathcal{D}}$. The complete prefix integral has zero derivative. The remaining tail shrinks where $a_\delta(1/2) \neq 0$, so $J_c' \to 0$. This fixes the constant. To justify differentiation despite the interior logarithmic zero, write $\ell(t) = \log|2\cos(\pi t)|$. At fixed $\nu > t_c$, the prefix integral is

$$\int_\delta^{\nu+\delta} \ell(t)\,dt + \int_{-\delta}^{\nu-\delta} \ell(t)\,dt.$$

Its derivative is $\ell(\nu+\delta) - \ell(\nu-\delta)$; the two terms at $\pm\delta$ cancel. The moving zero lies strictly inside the integration interval, while all its endpoints are regular. Thus this primitive differentiates normally on the strict branch. At $\nu = 1/2$ its derivative is zero. The tail integral has smooth integrand there, so the stated branch-entry limit follows.

It follows that $S_{\mathrm{L}}' = \log(\mathcal{H}_c/\mathcal{E}_b)$. Both positive quantities are larger roots with the same product $\mathcal{D}$:

$$\mathcal{E}_b = C_b + \sqrt{C_b^2 - \mathcal{D}}, \quad C_b = A + 2bz,$$
$$\mathcal{H}_c = C_c + \sqrt{C_c^2 - \mathcal{D}}, \quad C_c = 2cz - A.$$

Both $C_b, C_c \geq \sqrt{\mathcal{D}}$. The larger-root function is increasing, and hence

$$\operatorname{sign} S_{\mathrm{L}}' = \operatorname{sign}((c-b)z - A).$$

As z increases, this sign can pass only from negative to positive. There is no interior spectral maximum. At branch entry the value is the terminal value already bounded. At $\delta \to 1/2$, both $\mu_b, \nu_c \to 0$, and logarithmic integrability gives $S_{\mathrm{L}} \to \log c$. The upper bound follows.

The continuum candidates are attained: $\mu = \nu = 0$ gives $\log c$, and $\delta = 0, \nu = 1/2, \mu = \mu_b$ gives $\mathcal{Q}(x, b, y) + \frac{1}{2}\log c$. For $b = 0$, use its sole start or the continuous limiting formulas. For $P = 0$, the exact one-step root gives the same identities without division by P. The exceptional $\mathcal{D} = 0$ occurs only at $x = y, \delta = 1/2$, where the computed continuous endpoint values apply. □

7 Face phases and the last spectral cap

To use the signed real transfer at arbitrary face phases, we need positivity of the interval continuant itself, rather than positivity of its recurrence coefficients.

Lemma 7.1 (Continuous real positivity of interval tails)**.** *For $0 \leq i \leq j \leq m$ and $0 \leq \delta \leq 1/2 - 1/(2N)$, $K_{i,j}$ has nonnegative coefficients in x, y. Consequently*

$$|K_{i,j}(X_0, Y_0; \delta)| \leq K_{i,j}(|X_0|, |Y_0|; \delta).$$

Proof. Set $U_r = K_{i,i+r} / \prod_{v=i+1}^{i+r} s_v^2$. Then $U_0 = 1, U_1 = x + y$, and for $r \geq 2$, $j = i + r$,

$$U_r = (x+y)U_{r-1} + \beta_j xy U_{r-2}, \qquad \beta_j = \frac{\alpha_{j-1}}{s_j s_{j-1}}.$$

The recurrence starts at its second step, so it never divides by s_0. Product-to-sum gives the row-independent identity

$$\alpha_{j-1} + s_j s_{j-1} = 4\cos^2(\pi\delta) - 4\sin^2(\theta/2) \geq 0.$$

Thus $1 + \beta_j \geq 0$, even if $\beta_j < 0$. Simultaneous induction using

$$U_r - xU_{r-1} = y(U_{r-1} - xU_{r-2}) + (1 + \beta_j)xyU_{r-2}$$

and $U_0 = 1, U_1 - xU_0 = y$ proves positivity. Taking moduli in a polynomial with nonnegative coefficients proves the last assertion. □

In (6.1), take absolute values of the middle monomials and of the scalar unused products. The lemma then permits the positive outer moduli in the *same signed* continuant. Thus Proposition 6.1 bounds every face phase uniformly for

$$-2 \le \sigma \le b_N := 2\cos(\pi/N).$$

No positivity of the full numerator, or comparison of its unused products to their endpoint values, has been asserted.

For the small remaining interval, let p be any complex polynomial of degree at most d, bounded by M on $[-2, b_N]$. The affine coordinate of $\sigma = 2$ relative to this interval is $z_N = (6 - b_N)/(b_N + 2)$. For a degree-$d$ polynomial r, the function $w^{-d} r((w + w^{-1})/2)$ is holomorphic outside the unit disk and at infinity. On its boundary it is bounded by $\|r\|_{[-1,1]}$. The maximum principle on the reciprocal disk gives

$$|r(z)| \le \|r\|_{[-1,1]} \exp(d \operatorname{arcosh} z), \qquad z \ge 1.$$

Complex coefficients cause no change. As $z_N = 1 + O(N^{-2})$, uniformly for $N \ge 3$,

$$\sup_{-2 \le \sigma \le 2} |p(\sigma)| \le M \exp(d \operatorname{arcosh} z_N) \le CM. \tag{7.1}$$

For each fixed face-phase tuple the actual left polynomial has spectral degree at most d. Apply (7.1) before taking the supremum of those phases. The constant is independent of them. This proves the left assertion of Theorem 1.1. Row reflection proves the right assertion. The central assertion was proved in Section 5. Every compact estimate respects the declared zero pattern, completing the theorem.

8 Shared face coefficients

A fixed finite collection of boxes may be assigned genuine labels s_e, with every occurrence of each face variable identified. Keep designated zero faces exactly zero. Let t_f be the ordinary coefficient targets, and choose a single positive radius ρ_f for each remaining face. Require that the sum of the two outer moduli is positive in every box; a zero pattern killing both outer variables is outside Theorem 1.1. Cauchy's coefficient formula gives

$$\left| [L^t] \prod_e H_{V,s_e}(L; \sigma_e) \right| \le \prod_f \rho_f^{-t_f} \sup_{|L_f| = \rho_f} \prod_e |H_{V,s_e}(L; \sigma_e)|.$$

Theorem 1.1 applies at this shared radius vector. If $t_f/N \to \tau_f$, the uniform real-cube upper rate is at most

$$\sum_e A_{s_e}(\rho) - \sum_f \tau_f \log \rho_f.$$

The number of boxes is fixed before N grows, so their errors sum to $o(N)$. For finitely many sector strings, take the maximum of these sums before bounding their finite sum. This keeps correlations between genuine labels; no Cartesian enlargement is needed.

This argument gives an upper bound for each prescribed sector choice. The coefficientwise comparison (3.3) applies to the central sector. The one-sided estimates remain bounds on values; we have not proved coefficientwise domination for them.

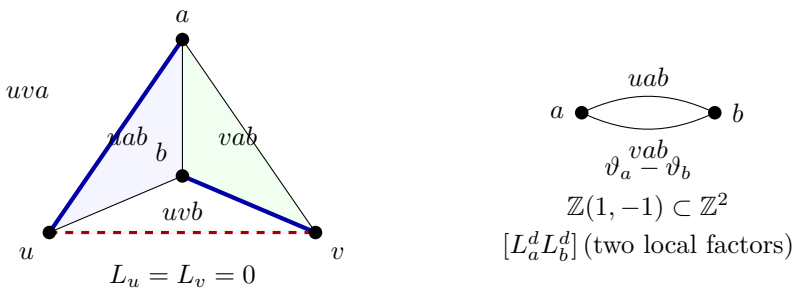


Figure 2: A combinatorial tetrahedral sphere and its ordinary phase graph. The vertices u, v, a, b name shared face variables. The marked edge uv (red, dashed) sets $L_u = L_v = 0$; the two blue matching edges are ua and vb. The outer face of the planar drawing is the triangle uva. The two triangles with two ordinary vertices, uab and vab, compare the same ordinary phases ϑ_a, ϑ_b, giving two copies of the primitive difference row $(1, -1)$. Their coefficients add in the Hessian, while the quotient angular measure remains $d\vartheta_b/(2\pi)$ after $\vartheta_a = 0$. The coefficient target is $L_a^d L_b^d$, with each variable shared by the two local factors. The edge lengths in this combinatorial drawing do not prescribe the radii in the variational problem.

8.1 The marked triangulation and its central coefficient

Let $\mathcal{T}$ be a finite simple oriented triangulation of the sphere. A perfect matching of its dual pairs its triangles across R edges. Its $R + 2$ vertices are called faces, following their role in the original diagram. Choose a nonmatching edge uv as the mark. The two face variables L_u, L_v are set to zero; the remaining R variables are ordinary. The order at a matched pair is (x, b, c, y): its triangles are (x, b, y) and (x, c, y), with their induced opposite orientations along the matching edge xy. Repeated occurrences of a face always denote the same variable. In particular $x + y > 0$ on the positive ordinary radii of every pair: otherwise its matching edge would be the unique edge uv, which is the nonmatching mark. Each pair also has a nonmonomial triangle. If one outer face is protected, only one other zero remains, so at least one middle is positive; if both outers are positive, both triangles are nonmonomial.

Use the operator and face polynomial of Section 2, with $d = N - 1$ and $J = \lfloor d/2 \rfloor$. The central sector is equivalently the set of paths with $i \le J$ and $d - j \le J$. At each matched edge we select this sector of the same divided difference V_N.

Put $\sigma_e = -2\cos(2\pi\delta_e)$ and define the raw central coefficient

$$C_N(\delta) = \left[\prod_{f \neq u,v} L_f^d\right] \prod_{e=1}^{R} H_{V,\mathrm{C}}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e}; \sigma_e)\Big|_{L_u = L_v = 0}. \tag{8.1}$$

This definition includes its sign $(-1)^{dR}$. Its identification as a contribution to the original marked evaluation, with cups, caps and framing still present, is the exact diagram theorem [1, Theorem 9.1 and Corollary 9.2]. The coefficient itself is the object of the present analytic theorem.

Here is the sole positive-radius input. It can be supplied by the complete packing of the marked triangulation, but is stated analytically so that the theorem can be used without a geometric normalization. Use the triangle function $\mathcal{Q}$ defined in (1.1), with every protected

face kept exactly zero. Choose positive ordinary radii ρ_f such that

$$\mathcal{F}(L) = \sum_{e=1}^{R} \{\mathcal{Q}(x_e, b_e, y_e) + \mathcal{Q}(x_e, c_e, y_e)\} - \sum_{f \neq u,v} \log L_f \quad \text{is stationary at } L = \rho \tag{8.2}$$

modulo common scale. Set $\Phi = \mathcal{F}(\rho)$. Zeros remain exactly zero. The diagram and radii are fixed before N tends to infinity.

Theorem 8.1 (Asymptotic of the shared central coefficient)**.** *Under these hypotheses there is a fixed complex polydisk about zero and, for all sufficiently large N, holomorphic functions A_N and holomorphic $(R-1)$-by-$(R-1)$ matrices H_N on it such that*

$$C_N(\delta) = \frac{(-1)^{dR} N^{2R} e^{N A_N(\delta)}}{(2\pi N)^{(R-1)/2} \sqrt{\det H_N(\delta)}} (1 + O(N^{-1})). \tag{8.3}$$

The relative error is holomorphic and uniform on smaller closed polydisks. The square root continues the positive face Hessian at zero. The actions are separately even and real on the real chart, and $A_N \to A_0$, $H_N \to H_0$ locally uniformly with every fixed derivative. The face Hessian $H_0(0)$ is positive definite. Moreover

$$\begin{aligned} A_0(0) = \Phi, \qquad A_0''(0) = -2\,\mathrm{diag}(\kappa_e), \\ \kappa_e = \frac{\pi(\sqrt{b_e(x_e+y_e)+x_e y_e} + \sqrt{c_e(x_e+y_e)+x_e y_e})}{x_e + y_e} > 0. \end{aligned} \tag{8.4}$$

All face values in the last expression are at ρ.

For every $0 < \eta < 1/2$ there are $\gamma_\eta, \tau_\eta > 0$ such that, for all sufficiently large integer levels,

$$|C_N(x + \mathrm{i}y)| \le e^{N(\Phi - \gamma_\eta)} \quad (x \in [-1/2, 1/2]^R,\ \max_e |x_e| \ge \eta,\ \max_e |y_e| \le \tau_\eta). \tag{8.5}$$

Fixed derivatives have the corresponding bounds on smaller domains. There is no prime-level restriction and no change of normalization between even and odd levels.

We define the finite actions directly from the actual local polynomial. They are not asserted to equal a separately constructed approximate fold interpolation. In particular no value-only interpolation is differentiated to obtain a normal.

9 Positive matrices at the finite spectral nodes

We use the cofactors R_j and interval polynomials $K_{i,j}$ of (2.5)–(2.6). Throughout the remaining sections, $u(\delta) = -4\sin^2(\pi\delta)$ denotes the spectral coordinate; it is distinct from the protected vertex u of the triangulation. We also write $P_{\mathrm{C}} = \mathcal{P}_{\mathrm{C}} = (-1)^d H_{V,\mathrm{C}}$. The central factorizations (2.7)–(2.8) retain the odd skipped-row term, including its derivative at $u = 0$.

We give the finite positive representation needed for phase estimates. Let $S_j = \prod_{r=1}^{j} s_r$, $S_0 = 1$, and, for $0 \le a \le J$, let $H_{\pm}^{(a)}$ be the Hermitian tridiagonal matrices with

$$\begin{aligned} (H_\pm)_{n,n} &= \frac{1}{\cos(n\theta)\cos((n+1)\theta)}, && 0 \le n < a, \\ (H_\pm)_{n-1,n} &= \sqrt{b_n}\{\sin(n\theta) \pm \mathrm{i}\cos(n\theta)\}, && 1 \le n < a, \\ b_n &= \frac{\sin^2(n\theta)}{\cos^2(n\theta)\sin((2n-1)\theta)\sin((2n+1)\theta)}. \end{aligned} \tag{9.1}$$

Empty matrices have determinant one. Put

$$\mathcal{D}_a(x,b,y) = \det[bI + xH_-^{(a)}/2 + yH_+^{(a)}/2]. \tag{9.2}$$

Lemma 9.1 (Exact node adapter)**.** *For $u_a = -a_a$ and $a \le j \le J$,*

$$R_a(b,u_a) = S_{2a}\mathcal{D}_a(x,b,y), \qquad R_j(b,u_a) = K_{a,j}(u_a)R_a(b,u_a). \tag{9.3}$$

Both $H_\pm^{(a)}$ are positive definite. The polynomials $\mathcal{D}_a$ and $K_{a,j}(u_a)$ have nonnegative face coefficients.

Proof. At u_a, the cofactor column on rows $0,\dots,a$ is a null vector of the original triangular matrix. If its coefficients are r_i, its row equations are

$$(a_i - a_a)r_i = s_{i+1}^2 A r_{i+1} + s_{i+2}s_{i+1}P r_{i+2}.$$

The leading coefficient is $\prod_{r<a}(a_r - a_a) = S_{2a}$, by $a_r - a_a = 4\sin((a+r+1)\theta)\sin((a-r)\theta)$. Define $\partial_q b^j = s_j b^{j-1}$. Since $a_i - a_a = s_a s_{a+1} - s_i s_{i+1}$, the normalized monic polynomial is the unique degree-a eigenpolynomial of

$$\mathcal{L} = \partial_q((b^2 + Ab + P)\partial_q), \qquad \lambda_a = s_a s_{a+1}.$$

The eigenvalues through J are distinct: their successive differences are $4\sin(2a\theta)\sin\theta > 0$.

For completeness its monic eigenpolynomials p_a obey

$$p_a = \left(b + \frac{A}{2\cos((a-1)\theta)\cos(a\theta)}\right)p_{a-1} - \frac{b_{a-1}}{4}\{A^2 - 4P\cos^2((a-1)\theta)\}p_{a-2}. \tag{9.4}$$

Here $p_0 = 1, p_1 = b + A/(2\cos\theta)$. One elementary derivation also checks that no lower-degree terms were dropped. Normalize $[j] = s_j/s_1$ and define moments by

$$\mu_0 = 1, \quad \mu_{-1} = 0, \qquad [r+1]\mu_r + A[r]\mu_{r-1} + P[r-1]\mu_{r-2} = 0 \quad (1 \le r \le 2J-1).$$

The sine addition formulas make $\mathcal{L}$ symmetric for $\langle b^i, b^j\rangle = \mu_{i+j}$. Distinct eigenvalues therefore make the p_a orthogonal on the available degrees. Multiplication by b is symmetric as well, so $bp_{a-1} - p_a$ has only the two previous components. Comparison of the two highest coefficients, using the displayed row equations, gives (9.4). Their norm recurrence has its nonzero second coefficient for generic (A,P), starting from norm one. This justifies the induction before polynomial continuation to all A,P; no positive moment assumption is used. All denominators have index less than N. The tridiagonal determinant (9.2) has precisely (9.4), proving the first identity.

The Cholesky pivots of either H are

$$p_{n-1}^H = \frac{\sin(n\theta)}{\cos(n\theta)\sin((2n-1)\theta)} > 0, \qquad 1 \le n \le a.$$

Writing $d_n = (H_\pm)_{nn}$ and $h_n = (H_\pm)_{n-1,n}$, substitution in $d_n - |h_n|^2/p_{n-1}^H = p_n^H$ verifies the factorization, including the last physical row. Thus both matrices are positive. Cauchy–Binet applied to their Cholesky factors proves coefficient positivity of (9.2). The second identity and coefficient positivity of $K_{a,j}(u_a)$ were proved in Lemma 3.1: after row a the forcing vanishes, and the solutions of the normalized recurrence (3.1) have nonnegative face coefficients despite its signed second term. □

At the maximal physical fold the scalar can be read without any interpolation. For $\delta_* = 0$ at odd levels and $\delta_* = 1/(2N)$ at even levels, (2.7)–(9.3) give

$$P_{\mathrm{C}}(\delta_*) = N^2\beta_J^2\mathcal{D}_J(x,b,y)\mathcal{D}_J(x,c,y)(x+y)^{d-2J}, \qquad \beta_J = S_{2J}/S_J^2. \tag{9.5}$$

Indeed $S_{N-1} = N$, $S_J^2 = N$ at odd levels; at even levels $S_{N-2} = N/(2\sin\theta)$, $S_J^2 = N/2$, and $s_m = 2$. Thus both the even forced factor and the odd simple node are literal.

10 Fixed complex cofactor neighborhoods

We next prove the local analytic input for the actual polynomial, independently of its nodal samples. Positive face entries range in a fixed compact set; each zero entry stays zero. Write $n = N/2 + O(1)$, $n \le \lfloor N/2 \rfloor$, for any terminal row used in (2.7)–(2.8). Throughout this section $x + y > 0$, as proved for every actual pair in Section 8. The entropy in (1.1) also allows the different singleton $(0, b, 0)$, but that singleton is not an admissible cofactor chart here.

Proposition 10.1 (Actual local logarithms)**.** *Assume $x, y, b \ge 0$ and $x + y > 0$ at the real reference. On a fixed sufficiently small complex u disk and fixed complex logarithmic neighborhood of the positive faces, every nonzero $R_n(b, u)$ is zero-free for all large N. Its logarithm, continued from positive faces at $u = 0$, satisfies*

$$N^{-1} \operatorname{Log} R_n(b, u) \longrightarrow \psi_b(u) \tag{10.1}$$

locally uniformly, with every fixed parameter derivative. The limit is

$$\begin{aligned} \psi_b(u) &= \phi(t_b(u); u), \\ \phi(t; u) &= t \operatorname{Log} b + \int_0^t \operatorname{Log}(4 \cos^2 \pi v + u)\, dv + \int_t^{1/2} \operatorname{Log} \lambda(v, u)\, dv, \\ \lambda(v, u) &= 2 \sin \pi v \left[A \sin \pi v + \sqrt{(x - y)^2 \sin^2 \pi v + (4 + u) P} \right], \\ \sin^2 \pi t_b(u) &= \frac{(1 + u/4) b^2}{(b + x)(b + y)}. \end{aligned} \tag{10.2}$$

For $b = 0$, the expression means $\int_0^{1/2} \operatorname{Log} \lambda$; the sole-variable case is the exact monomer product. All branches continue their positive values at $u = 0$.

10.1 An interior starting row

Suppose $b > 0$. The starting saddle $t_b(0)$ lies strictly between zero and $1/2$. Choose $0 < \ell < t_b(0) < r < 1/2$ with fixed margins, uniformly over the face compact. On a complex neighborhood of $[\ell, 1/2]$, the recurrence for K has coefficients

$$p(t) = 4 \sin^2(\pi t) A, \qquad q(t, h) = 4 \sin(\pi t) \sin(\pi (t - h)) [4 \cos^2(\pi (t - h/2)) + u] P, \quad h = N^{-1}.$$

Its dominant characteristic root is a unit, and the ratio of the other root to it has modulus at most $\kappa < 1$. This remains true on fixed sufficiently small complex parameter neighborhoods. A zero upper diagonal can annihilate the smaller root and causes no problem. If $P = 0$ the recurrence is first order.

Here are the needed analytic error estimates. Starting with the characteristic root, iterate

$$r_{k+1}(t) = p(t) + q(t, h)/r_k(t - h)$$

on nested rectangles, advancing their left margins by h. On an $O(h)$ neighborhood of the root, translation changes the root by $O(h)$ and the derivative in the denominator has modulus at most $\kappa' < 1$. After $\lfloor cN \rfloor$ steps, with c smaller than the original margin, the resulting analytic unit r_N satisfies

$$r_N(t) - p(t) - q(t, h)/r_N(t - h) = O(e^{-c_1 N}), \qquad r_N(t) = \lambda(t, u) + O(h). \tag{10.3}$$

One extra shifted rectangle is retained for the residual. The contraction is between nested domains; no translation-invariant analytic norm is assumed. The constants and Cauchy derivative bounds are uniform.

Divide the actual $K_{i,j}$ by $\prod_{v=i+1}^{j} r_N(vh)$. In the zero-residual recurrence its consecutive differences multiply by $-k_N(jh)$, where $k_N(t) = q(t,h)/(r_N(t)r_N(t-h))$ has modulus at most κ'. The actual initial data $K_{i,i} = 1, K_{i,i-1} = 0$ consequently give

$$K_{i,n} = a_N(ih) \prod_{v=i+1}^{n} r_N(vh)(1 + O(e^{-c_2 N})), \tag{10.4}$$

$$a_N(t) = \sum_{l=0}^{\lfloor c_3 N \rfloor} (-1)^l \prod_{v=1}^{l} k_N(t + vh). \tag{10.5}$$

The omitted geometric tail is exponential; a_N tends to $(1 + k_0(t))^{-1}$, a positive unit. The residual in (10.3) adds $O(Ne^{-c_1 N})$ by the same difference recurrence. This proves (10.4) for every actual start $i/N \in [\ell, r]$, with its physical terminal row.

Both logarithmic products in (2.6) admit exponentially accurate analytic primitives on this starting interval. For clarity, the Euler–Maclaurin formula with its periodic Bernoulli remainder [3, §2.10(i)] gives, after $2K$ derivatives, the bound

$$Ch^{2K-1} \frac{(2K)!}{(2\pi d_0)^{2K}} \le Ch^{-1}(C_0 K h)^{2K}$$

on a retained strip of width d_0. The midpoint formula obeys the same estimate. It follows from the periodic Bernoulli remainder and Cauchy's derivative bound; choose $K = \lfloor c_0/h \rfloor$ with $C_0 c_0 < 1$. Anchor the prefix at zero and the tail at the actual n/N. Thus each starting term is $a_N(i/N)e^{N\phi_N(i/N)}(1 + O(e^{-c_4 N}))$, with analytic finite ϕ_N converging with derivatives to (10.2). All endpoint half terms are retained in ϕ_N.

The omitted starts have a fixed rate loss at positive faces and $u = 0$. Indeed Lemmas 4.1 and 5.2 give the uniform interval upper bound for (2.6), and its action has its unique strict maximum at $t_b(0)$. This loss persists on a fixed small complex parameter neighborhood. One direct quantitative check is to bound each unused diagonal by $a_j + |u|$. The product over distinct used lower rows costs at most $\exp\{N\omega(|u|) + O(\log N)\}$, with $\omega(v) \to 0$: split rows where $a_j \le \sqrt{v}$ (at most $O(Nv^{1/4}) + O(1)$), and use $\log(1 + v/a_j) \le \sqrt{v}$ on the others; the endpoint log sum on the first set is integrable and $O(Nv^{1/4}|\log v|) + O(\log N)$. The odd cofactor never uses its central zero diagonal. A logarithmic face perturbation costs $e^{C\rho N}$ by homogeneous degree. Choose these neighborhoods smaller than the fixed starting-row loss.

We record why the discrete saddle is a contributing one. Surround the kept integer grid by a rectangle with vertical sides on half-grid lines. For $w_N(t) = a_N(t)e^{N\phi_N(t)}$, residues give exactly

$$\sum_i w_N(i/N) = \frac{N}{2\mathrm{i}} \oint \cot(\pi N z) w_N(z)\, dz. \tag{10.6}$$

On the upper and lower sides at fixed height η the cotangent is $\mp\mathrm{i} + O(e^{-2\pi\eta N})$; on the vertical sides its modulus is at most one. The latter retain the endpoint rate loss. The horizontal phase cost at the real reference is $O(\eta^2 N)$, so choose η first with $2\pi\eta > C\eta^2$, then choose the parameter neighborhood. Cauchy's theorem compares (10.6) with N times the actual integral, retaining exponentially smaller vertical currents. The holomorphic implicit function theorem gives its saddle t_N. Deforming to the horizontal line through t_N,

with the same endpoint currents, and rescaling by $N^{-1/2}$ yields

$$R_n = \frac{\sqrt{2\pi N}\, a_N(t_N) e^{N\phi_N(t_N)}}{\sqrt{-\phi_N''(t_N)}}(1 + O(N^{-1})). \tag{10.7}$$

The cubic Gaussian term integrates to zero; its square and the quartic term are $O(N^{-1})$. Exponential interpolation errors remain relative errors after the neighborhood is decreased once more. Uniform holomorphy gives the fixed derivative bounds. This proves zero exclusion and (10.1) for $b > 0$.

10.2 A protected middle variable

If $b = 0$ and $xy > 0$, only start zero remains, so it must not be discarded by the preceding interior argument. Set $L_j = R_j/S_j$. Exactly,

$$\begin{pmatrix} L_j \\ L_{j-1} \end{pmatrix} = T_h(jh)\begin{pmatrix} L_{j-1} \\ L_{j-2} \end{pmatrix}, \qquad T_h(t) = \begin{pmatrix} 2\sin(\pi t)A & P[4\cos^2(\pi(t-h/2)) + u] \\ 1 & 0 \end{pmatrix}.$$

The initial vector is $(1,0)^{\mathsf{T}}$. Let $\lambda_\pm(t)$ be the two roots and $g = \lambda_+ - \lambda_-$. The real discriminant at $h = u = 0$ is $4\sin^2(\pi t)A^2 + 16\cos^2(\pi t)P > 0$ on $[0, 1/2]$. Thus g and λ_+ are units on a fixed complex neighborhood. The ratio $r = \lambda_-/\lambda_+$ equals -1 at zero, while $r'(0) = 2\pi A/\lambda_+(0)$ has positive real part. By compactness,

$$|r(t)| \le e^{-ct} \quad (0 \le t \le 1/2), \qquad |1 - r(t)| \ge c.$$

The equality of moduli at the initial point is not an eigenvalue collision.

In the eigenbasis $E(t) = \left(\begin{smallmatrix} \lambda_+ & \lambda_- \\ 1 & 1 \end{smallmatrix}\right)$, let $B_j = E(jh)^{-1}E((j-1)h)$ and $C_j = (B_j)_{++} = (\lambda_+((j-1)h) - \lambda_-(jh))/g(jh)$. Divide out $\prod_{j=1}^n \lambda_+(jh)C_j$. The update is $\operatorname{diag}(1, d_j) + F_j$, with off-diagonal $F_j = O(h)$, $F_j' = O(h)$, $|1 - d_j| \ge c$, and $|d_j| \le e^{Ch - cjh}$. Conjugate by $I + X_j$, where its two off-diagonal entries solve $\operatorname{diag}(1, d_j)X_j - X_j \operatorname{diag}(1, d_j) = -F_j$. Then $X_j = O(h)$ and $X_j - X_{j-1} = O(h^2)$, so the new error is $O(h^2)$. All partial products are bounded by e^{Cnh}; telescoping gives an $O(h)$ total error. The negative diagonal product is $O(e^{-c'N})$ for n/N bounded away from zero. The initial eigenbasis coordinates are exactly $(1/g(0), -1/g(0))$. Inverting the final basis proves

$$R_n = \frac{\lambda_+(nh)}{g(0)} \prod_{j=1}^{n} [s_j \lambda_+(jh) C_j](1 + O(N^{-1})). \tag{10.8}$$

Its amplitude is a uniformly bounded unit, since $C_j = 1 + O(h)$. Riemann sums, with the integrable initial $\log\sin$ term, prove (10.1); the analytic relative estimate gives all fixed derivatives. If $b = 0, xy = 0$, the exact answer is $R_n = A^n \prod_{j=1}^n s_j^2$, independent of u. This completes Proposition 10.1.

10.3 The exact central interpolant and its action

The terminal ratios in both proofs tend uniformly to

$$\frac{R_n}{R_{n-1}} \longrightarrow \lambda_* = 2\{A + \sqrt{A^2 + uP}\}.$$

For interior starts the terminal forcing is separated from their saddle and exponentially smaller; for a protected start it follows directly from (10.8). Consequently the exact central factors satisfy

$$\frac{P_{\mathrm{C}}}{D_b D_c} \longrightarrow 4\sqrt{A^2+uP}(N \text{ even}), \qquad \frac{P_{\mathrm{C}}}{G_b G_c} \longrightarrow \frac{2\sqrt{A^2+uP}}{A+\sqrt{A^2+uP}}(N \text{ odd}). \tag{10.9}$$

These are units near zero, including one-outer-zero cases. Define the exact holomorphic interpolation

$$\mathcal{B}_N(\delta; L) = N^{-2} P_{\mathrm{C}}(u(\delta); L). \tag{10.10}$$

It is zero-free on a fixed chart, is separately even in δ, and is positive on its sufficiently small real positive-face chart. Its physical maximal-fold value is exactly $(9.5)/N^2$. Its logarithm is therefore fixed without a fitted scalar. Equations (10.1) and (10.9) show

$$N^{-1} \operatorname{Log} \mathcal{B}_N(\delta; L) \longrightarrow Q_b(\delta; L) + Q_c(\delta; L), \tag{10.11}$$

with fixed derivatives. Lemma 5.1 gives the absolute normalization and derivative

$$\begin{aligned} Q_b(0) &= \mathcal{Q}(x, b, y), \\ Q_b'(\delta) &= -\operatorname{Log} \frac{\mathcal{E}_b(\delta)}{\sqrt{\mathcal{D}(\delta)}}, \quad \mathcal{D} = A^2 - 4P \sin^2 \pi\delta, \\ \mathcal{E}_b(\delta) &= A + 2b \sin^2 \pi\delta + 2 \sin \pi\delta \sqrt{bA + P + b^2 \sin^2 \pi\delta}. \end{aligned} \tag{10.12}$$

That lemma proves its additive constant directly from the real cofactor action, using the entropy derivatives and the $b \to \infty$ limit; it makes no volume normalization. Its formula on a real interval equals the limit (10.2), and the identity theorem therefore supplies the same holomorphic branch in (10.11). The constant triangle has $Q_b = \frac{1}{2} \log A$.

11 All face phases and the primitive Gaussian

We prove a global angular bound, not just positive real convexity. For positive semidefinite matrices X_j with $M = \sum X_j > 0$, put $A_j = M^{-1/2} X_j M^{-1/2}$ and $B = \sum e^{\mathrm{i}\vartheta_j} A_j$. The map $z \mapsto (A_j^{1/2} z)_j$ is an isometry, so B is a contraction. Expanding its trace gives

$$|\det B| \le \exp\left[-\frac{1}{2} \sum_{j<k} |e^{\mathrm{i}\vartheta_j} - e^{\mathrm{i}\vartheta_k}|^2 \operatorname{Tr}(A_j A_k) \right]. \tag{11.1}$$

Indeed $\operatorname{Tr}(I - B^* B)$ is the sum displayed without $1/2$, and $\log t \le t - 1$ for the squared singular values. Zero determinants follow by continuity. No commutativity is required.

Lemma 11.1 (Finite triangle phase loss)**.** *For every $a_0 > 0$ and compact positive nonzero-face set, a triangle with at least two positive entries has a phase loss e^{-cN} in $\mathcal{D}_a$ whenever their circular phase diameter is separated from zero, uniformly for $a_0 N \le a \le J$.*

Proof. First let $b > 0$ and use $M = bI + xH_-/2 + yH_+/2$. For each positive outer summand X, its pair trace with bI is $b \operatorname{Tr}(X M^{-2})$. Select every other Cholesky square of H with indices in $[a_0 N/4, a_0 N/2]$. Their normalized vectors w_j have disjoint supports; their pivots and row norms are bounded above and below, and there are at least cN of them. Thus $X \ge c \sum w_j w_j^*$. On these supports $\langle w_j, M w_j \rangle \le C$. Jensen's inequality for the spectral measure of M gives $\langle w_j, M^{-2} w_j \rangle \ge C^{-2}$. Hence each applicable pair trace is at least

cN. At three positive entries, one phase difference from b is at least half their diameter. Equation (11.1) proves the claim.

If $b = 0$ and $x, y > 0$, the determinant has the exact expansion

$$\mathcal{D}_a(x, 0, y) = \beta_a^{-1} \sum_{k=0}^{a} \left(\frac{S_a}{S_k S_{a-k}} \right)^2 x^k y^{a-k}, \qquad \beta_a = S_{2a}/S_a^2. \tag{11.2}$$

It follows by substituting the coefficient ratios s_{a-k}^2/s_{k+1}^2 in the continuant (9.4), with its empty and one-row values. For its positive weights the adjacent ratio is $R(k) = (x/y)[\sin((a-k)\theta)/\sin((k+1)\theta)]^2$ and

$$(\log R)'(t) = \frac{-2\theta \sin((a+1)\theta)}{\sin((a-t)\theta)\sin((t+1)\theta)}.$$

The mode stays cN from both ends. This derivative is at most $-c/N$ everywhere and has absolute value at most C/N near the mode. Multiplication of the ratios bounds the weights above by a Gaussian of width $\sqrt{N}$, and below by a fixed fraction of the modal weight on both shells at distances between $\sqrt{N}$ and $2\sqrt{N}$. The total mass is comparable to $\sqrt{N}$ times the modal weight. Two independent samples from these shells therefore have variance at least cN. The same variance is the log-radius Hessian of (11.2), hence the pair trace in (11.1). This proves the remaining case. A one-variable triangle is a monomial and is explicitly exempt. □

Lemma 11.2 (Terminal cofactor phase loss)**.** *If two positive triangle faces have separated phases, every required R_{J-k}, with k fixed, has an exponential angular loss relative to its positive value at $u = 0$, uniformly for all real $u \in [-4, 0]$ and on a sufficiently thin fixed complex δ strip.*

Proof. We first bound its actual nodes u_a, $a \le J - k$. Choose a small fixed $\delta_0 > 0$. For $a/N \ge 1/2 - \delta_0$, use (9.3) and Lemma 11.1; the remaining K has nonnegative coefficients and its positive value is bounded by its all-monomer product. Both factors are bounded coefficientwise by their $u = 0$ counterparts, as in Lemma 3.1.

For the far nodes $a/N \le 1/2 - \delta_0$, there is a strict loss even without a phase restriction. If $xy > 0$, choose cN disjoint pairs of steps in $[1/2 - \delta_0/2, 1/2 - \delta_0/4]N$. At $u = 0$, replacing either selected monomer pair by its dimer multiplies its weight by at least a fixed positive constant. The independent choices show $K_{a,j}(0) \ge e^{c'N} \prod_{v=a+1}^{j} s_v^2 A$. But $K_{a,j}(u_a)$ is bounded by that monomer product. The product $K_{a,j}(0)R_a(0)$ is an actual subset of paths in $R_j(0)$, so the asserted nodal loss follows. If $xy = 0, b > 0$, write the zero-spectral starts as

$$W_i = b^i \prod_{r<i} a_r \prod_{v=i+1}^{j} s_v^2 A.$$

At a far node, starts $i > a$ vanish, and starts $i \ge \eta N$ cost $\prod_{r<i}(1 - a_a/a_r) \le (1-c)^{\eta N}$. For small fixed η, the exact ratio $W_{i+1}/W_i = ba_i/(s_{i+1}^2 A)$ is at least $1 + c$ for $i \le 2\eta N$. Every remaining short start is consequently exponentially below $W_{\lfloor 2\eta N \rfloor}$. This covers all nonmonomial zero patterns.

The spectral degree of R_j is at most j. The full cosine-node interpolation on $u_0, \dots, u_J$ has norm $O(\log N)$ by Lemma 3.2. If $j < J$, interpolate instead $R_j(u) \prod_{a=j+1}^{J}(u + a_a)$; its unavailable node values are exactly zero. Removing these k factors costs only a polynomial. Indeed a degree-j polynomial p has $\|p'\| \le C(j+1)^3\|p\|$ by its Chebyshev expansion. On a one-sided interval of length $c(j+1)^{-3}$ next to a maximum, $|p| \ge \|p\|/2$. Delete small

neighborhoods of the k roots, of less than half that length. At a remaining point every factor has modulus at least $c_k(j+1)^{-3}$. Division costs $C_k(j+1)^{3k}$. This works for complex coefficients and clustered roots. Thus the nodal loss extends to the whole real interval. Finally, for degree j,

$$|p(-4\sin^2\pi\delta)| \le e^{2\pi j|\Im\delta|}\|p\|_{[-4,0]}.$$

This is the exterior maximum principle applied to $w^{-j}p(-2-w-w^{-1})$, $|w| = e^{2\pi|\Im\delta|}$. A small fixed strip preserves half the loss. □

Delete the protected vertices u, v from the ordinary graph. It remains connected: reroute a path around u along its cyclic link, then around v along its link with u removed; the latter is a path or cycle and is connected. Every surviving edge is in a triangle with two ordinary positive entries. No matched edge is uv, so $x + y > 0$ in every box. Each box has at least one nonmonomial triangle: if an outer variable is zero, the other protected variable can kill at most one middle.

The active difference rows $e_f - e_g$ therefore contain a spanning tree of the R ordinary vertices. They span the integer lattice $\{k \in \mathbb{Z}^R : \sum k_f = 0\}$, not a proper sublattice. Fixing one face phase leaves an $(R-1)$-torus with no additional aliases. If its phase lies outside any neighborhood of zero, some active pair has a separated phase. Every term in (2.7)–(2.8) contains a terminal cofactor from each triangle. Lemma 11.2 therefore gives a loss in that term. All other factors obey the uniform coefficientwise central or cofactor bound from Proposition 3.3 and Lemma 3.2. The finitely many central prefactors compare to $P_{\mathrm{C}}(0)$: use $R_j(0) \ge s_j^2 A R_{j-1}(0)$ and $A^2 \ge 2P$. Consequently, uniformly for every real spectral vector and a thin strip,

$$\left|\prod_e P_{\mathrm{C},e}(\delta_e; \rho e^{\mathrm{i}\vartheta})\right| \le e^{-cN}\prod_e P_{\mathrm{C},e}(0;\rho) \tag{11.3}$$

on the complementary face torus. Polynomial factors have been absorbed by decreasing c. This proves an estimate for every central summand, not a choice of a favorable one.

12 The finite shared saddle

For the entropy (1.1), the maximizing loads at positive entries are

$$a_i = \pi^{-1}\arctan(w_i/D), \qquad D = \sqrt{w_1w_2 + w_2w_3 + w_3w_1}.$$

They follow from $e'(a) = \log\cot(\pi a)$ and the sum constraint; zeros are interpreted by continuity. Differentiating in logarithmic coordinates gives the positive graph form

$$d^2\mathcal{Q} = \sum_{i<j}\frac{w_iw_j}{2\pi D(w_i+w_j)}(d\log w_i - d\log w_j)^2. \tag{12.1}$$

Terms on zero coordinates are omitted; for a one-variable triangle the Hessian is zero. The ordinary graph connectivity makes the sum positive definite after fixing one ordinary scale. This is an explicit finite-dimensional inference from (8.2), not an assumption about a Gaussian cycle.

Fix one ordinary radius at its reference value and write $L_f = \rho_f e^{z_f}$ for the other $r = R - 1$ faces. On the fixed local chart define

$$\Psi_N(z,\delta) = \frac{1}{N}\sum_e \operatorname{Log}\mathcal{B}_{N,e}(\delta_e; L(z)) - \frac{d}{N}\sum_{f\ne u,v}\operatorname{Log}L_f(z). \tag{12.2}$$

All constants, including $O(\log N)/N$ terms and the even fold offset, remain here. Only the explicit N^{2R} is outside this phase. Proposition 10.1 gives convergence with derivatives to

$$\Psi(z,\delta)=\sum_e\{Q_{b_e}(\delta_e;L)+Q_{c_e}(\delta_e;L)\}-\sum_{f\neq u,v}\operatorname{Log}L_f.$$

At $(0,0)$ its z gradient vanishes by (8.2), and its Hessian is positive definite by (12.1). The quantitative holomorphic implicit function theorem now supplies unique nearby $z_N(\delta)$ on one fixed polydisk, converging with derivatives. For example, precondition the gradient by the inverse limiting Hessian: its derivative is uniformly close to the identity on a fixed ball, and its value at zero tends to zero; contraction on a smaller ball proves the assertion uniformly in δ. Set

$$A_N(\delta)=\Psi_N(z_N(\delta),\delta),\qquad H_N(\delta)=\partial_z^2\Psi_N(z_N(\delta),\delta).\tag{12.3}$$

Uniqueness preserves separate evenness and reality.

The coefficient in (8.1) has the exact Cauchy measure $(2\pi)^{-r}d\vartheta$ on this torus. Indeed the numerator has total degree dR, equal to the sum of its ordinary targets; integrating the common phase gives one. Choosing a reference vertex is a primitive integer change of torus variables, so no lattice index multiplies that measure. One may change the radii and phase origins to $\rho e^{z_N(\delta)}$: the integrand is a Laurent polynomial and crosses no pole in the annuli. Near the saddle it is exactly

$$(-1)^{dR}N^{2R}e^{NA_N(\delta)}\exp\left[-\frac{N}{2}\vartheta^{\mathsf T}H_N(\delta)\vartheta+O(N|\vartheta|^3)\right].$$

The real part of H_N is uniformly positive definite on a smaller spectral polydisk. The complementary face torus remains exponentially small by (11.3): its real radius changes only within the fixed positive compact, and its phase origin changes by a small amount. Decrease the spectral chart so this fixed loss exceeds the change in $\Re A_N$. Rescale $\vartheta=N^{-1/2}v$. A fixed small neighborhood is dominated by $e^{-c|v|^2}$; outside it the angular bound applies. Taylor expansion through fourth order, with uniform analytic derivative bounds, gives the Gaussian determinant. The cubic term is odd; its square and the quartic term have relative size $O(N^{-1})$. The remaining Taylor terms and Gaussian tails obey the same bound after a standard $|v|\le N^{1/10}$ split. This proves (8.3), including its coefficient one and its holomorphic relative error.

At zero, mixed z,δ derivatives vanish by separate evenness. Eliminating the face variables therefore does not change the spectral second derivative. Differentiating (10.12) at zero gives

$$Q_b''(0)=-2\pi\sqrt{bA+P}/A.$$

This proves (8.4). Each box has a nonmonomial triangle, so its constant is strictly positive.

13 The entire spectral interval and a fixed strip

The local spectral decrease uses the actual signed transfer, not coefficientwise domination by a positive endpoint. From Lemmas 4.1 and 5.2, for every terminal row $N/2+O(1)$,

$$\limsup\frac{1}{N}\log|R_j(b,u(\delta))|\le\max\{Q_b(\delta),\tfrac12\log b\}\quad(0\le\delta\le1/2).\tag{13.1}$$

For $b=0$ the second branch is omitted. The bound is uniform on the whole interval. By (10.12), Q_b decreases strictly when $b>0$ or $xy>0$. Also $Q_b(0)>\frac12\log b$ in that case: the

endpoint load is not the maximizer of the strictly concave entropy when another entry is positive. The possible endpoint logarithmic singularity at $x = y, \delta = 1/2$ is integrable. Thus a separated $|\delta| \geq \eta$ gives a fixed loss in every nonmonomial triangle.

We specify the small complex face extension used in sewing this bound. In the proof of Lemma 4.1, first remove a small initial interval, then choose a fixed block length so the Cayley–Hamilton bound for each two-dimensional transfer power is within any prescribed exponential error of its spectral radius. Only after choosing that length, perturb the face parameters. Matrix powers over this fixed length vary continuously, so the same block bound persists. The initial interval has the same integrable logarithmic allowance. The scalar unused products are unchanged by face phases. Hence (13.1), with any prescribed error, holds on a fixed sufficiently small complex face neighborhood. This argument neither inverts an eigenbasis at a repeated root nor assumes phase positivity.

Fix the original radii ρ. On the small common-face-phase neighborhood, one separated spectral coordinate supplies the above loss in its nonmonomial triangle; the other factors have their complete upper rates with errors whose finite sum is smaller than half that loss. Equations (2.7)–(2.8) retain both triangle cofactors in each term, and their remaining prefactors are bounded. On the complementary face torus use (11.3), uniform for every real spectral vector. The Cauchy target factor is the same $\prod \rho_f^{-d}$ in both regions. Its complete rate is precisely Φ from (8.2). We have therefore proved a real-cube gap for the shared coefficient, not a sum of separately optimized factors. The same argument without a loss, or the coefficientwise central comparison, gives

$$\sup_{\delta \in [-1/2,1/2]^R} |C_N(\delta)| \leq e^{N(\Phi + o(1))}. \tag{13.2}$$

For clarity a final elementary argument supplies a fixed complex strip directly for this shared polynomial. A Laurent polynomial F of degree at most d in each angular coordinate has, in either half-plane of each variable, the subharmonic estimate

$$\log |F(x + \mathrm{i}y)| \leq (\mathcal{P}_y * \log |F|)(x) + 2\pi d \sum_e |y_e|. \tag{13.3}$$

Here $\mathcal{P}_y$ is the product of the circle Poisson kernels at radii $e^{-2\pi|y_e|}$; a zero height denotes a point mass. To prove it, multiply by the appropriate z_e^d in each coordinate, apply the disk subharmonic inequality successively, and subtract the known monomial logarithms. Zeros follow by regularization, or by using a continuous upper majorant on the boundary.

Apply the real gap with separation $\eta/2$ and (13.2). If $|x_j| \geq \eta$, the one-coordinate Poisson mass in $|t_j| \geq \eta/2$ tends uniformly to one as $|y_j| \to 0$. Choose the strip so it is at least $3/4$, and so $2\pi R|y|$ and the $o(1)$ term together cost less than a quarter of the real loss. Inequality (13.3) proves (8.5), possibly with a smaller constant. Cauchy's formula on smaller disks and strips gives the fixed derivative assertions. This completes Theorem 8.1.

14 The normalized input to the filling theorem

The hypotheses of [2, Theorem 1.1] now hold for the actual sequence (8.1), with

$$\varepsilon_N = (-1)^{dR}, \qquad \alpha = 2R - (R-1)/2 = (3R+1)/2, \qquad a_N(\delta) = \frac{(2\pi)^{-(R-1)/2}}{\sqrt{\det H_N(\delta)}}. \tag{14.1}$$

The limiting a_N is a nonzero holomorphic function. The finite functions A_N must remain inside the exponential; their convergence does not justify replacing e^{NA_N} by e^{NA_0} in a relative formula.

For fixed independently signed sufficiently large powers n_e, that theorem applies to the original central Fourier period with scalar $\Gamma_N = \exp(\pi \mathrm{i} d^2 \sum n_e/(4N))$. The product of its parity factor and (14.1) is $(-1)^{(N+1)R+dR} = 1$ at every integer level. If z_N is the unit-charge spectral saddle, S_N its finite action, and Q_N its negative spectral Hessian, the resulting leading expression is

$$\Gamma_N(-2\mathrm{i})^R \sqrt{2\pi}\, N^{R+1/2} \frac{\prod_e \sin(2\pi z_{N,e})}{\sqrt{\det H_N(z_N)}\sqrt{\det Q_N}} e^{NS_N}.$$

The relative error and eventual nonvanishing are precisely the conclusions of the oscillatory theorem. Its proof supplies the whole charge sum and endpoint currents; none is inferred from the present face Gaussian alone. Exterior framing or component units, and the modified-dimension factor when that convention is used, remain those of [1].

A geometric companion may identify $2\pi\Phi$ with an augmented volume and $2\pi\Re S_0$ with a specified filled volume. Those statements require its marked peripheral and metric proofs. A full invariant limit also requires control of the complementary degree sectors. Neither conclusion is part of Theorem 8.1.

References

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